Evaluation of family planning policy in India using Synthetic Control methods

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Dec, 2025

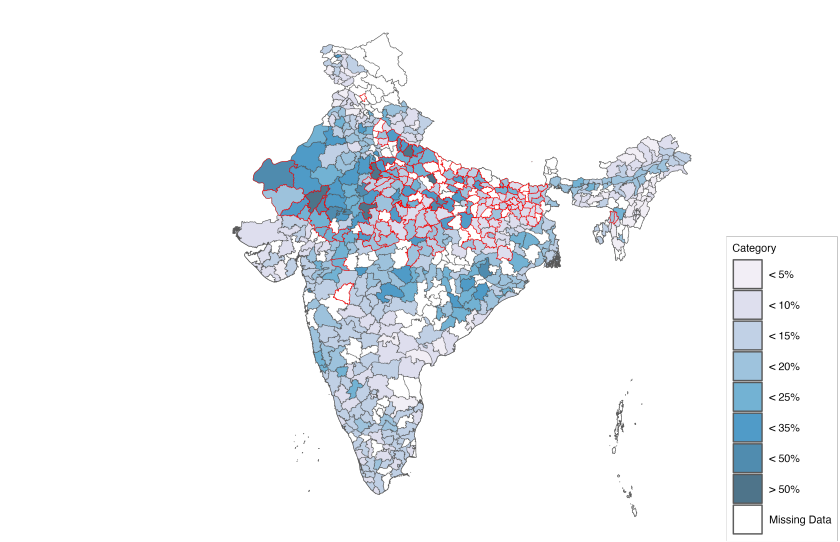
Research Question

Did the expansion of a family planning policy (*MPV*) impact the health outcomes of infants and mothers?

Motivation

- Policy aim was to increase modern contraceptive prevalence in high fertility districts that contribute to:
  - 28% of India's population
  - 30% of maternal deaths
  - 50% of infant deaths

Heat Maps: Fraction of LBW infants



What does MPV do?

- MPV
  - increase access to a range of ⏩ contraceptives
  - ⏩ monetary incentives for beneficiaries who adopt family planning methods
  - information dissemination using ⏩ mobile vans covering all blocks of a district

Literature Review

- Literature on the effect of increasing supply and access to family planning (Luca et al. (2021); Pfeifer and Stockburger (2023); Almanza and Rueda (2020); Anukriti et al. (2022); Paxton (2017)) have a common theme that both monetary and non-monetary costs that women face in access is important for determining take-up and health outcomes
- Current literature on MPV (Agarwal et. al. - WP) suggestive of decreasing fertility and increasing modern contraceptive prevalence rate

This Paper

- Documents the effects of MPV on infant and maternal outcomes
- Adopts Generalized Synthetic Control Method (GSCM) to estimate the ATT
  ⏩ Gsynth.vs.DiD
- Key findings:
  - Deliveries ↗ in the treated districts by 4.6 per 1000 women of reproductive age (15-49) with 3.9 infants born in health facilities
  - Heterogeneous impact on districts with:
    - high economic activity show ↘ in fraction of LBW (low birthweight) infants
    - low economic activity show ↗ in fraction of LBW (low birthweight) infants

Data

- Health Management Information System (HMIS) of India
  - Panel dataset of districts restricted for years 2009-19 for health indicators
- Census (2011)
  - Women of reproductive age (15-49)
- Socioeconomic High-resolution Rural-Urban Geographic (SHRUG) platform
  - Panel dataset of districts for years 2009-19 for night light data

Summary Statistics

Research Design

- Estimating equation:

$$Y_{it} = \delta_{it} D_{it} + \beta X_{it} + \lambda_i' f_t + \epsilon_{it} \tag{1}$$

  - D_{it}: "switches on" if district i is treated by MPV program in 2016
  - δ_{it}: effect of MPV on unit i at time t
  - X_{it}: vector of observed covariates
  - f_t: vector of unobserved common factors
  - λ_i: vector of factor loadings

Generalized Synthetic Control Method

- Define Y_{it}^I and Y_{it}^N as potential response with and without the intervention respectively

$$\delta_{it} = Y_{it}^I - Y_{it}^N, \qquad t > T_0 \tag{2}$$

- Generalized Synthetic Control Method (Xu, 2017)

$$\hat{Y}_{jt}^N = \theta_t' \underbrace{X_j}_{\text{observed}} + \underbrace{\lambda_j'}_{\text{unobserved FL}} \underbrace{f_t}_{\text{unobserved CFs}} + \underbrace{\epsilon_{jt}}_{\text{error term}} \tag{3}$$

$$\hat{\delta}_{it} = Y_{it}^I - \hat{Y}_{it}^N, \qquad t > T_0 \tag{4}$$

Results

ATT ESTIMATES OF HEALTH OUTCOMES OF INFANTS

Dependent Variable	Deliveries	LBW	Weighed	Fraction LBW	Institutional Deliveries
	(1)	(2)	(3)	(4)	(5)
ATT	0.0046***	-0.0006	0.0032**	-0.0015	0.0039**
	(0.0014)	(0.0010)	(0.0016)	(0.0145)	(0.0017)

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01. **LBW** = Low-Birth Weight infants

Heterogeneous Treatment Effects - Infant Outcomes

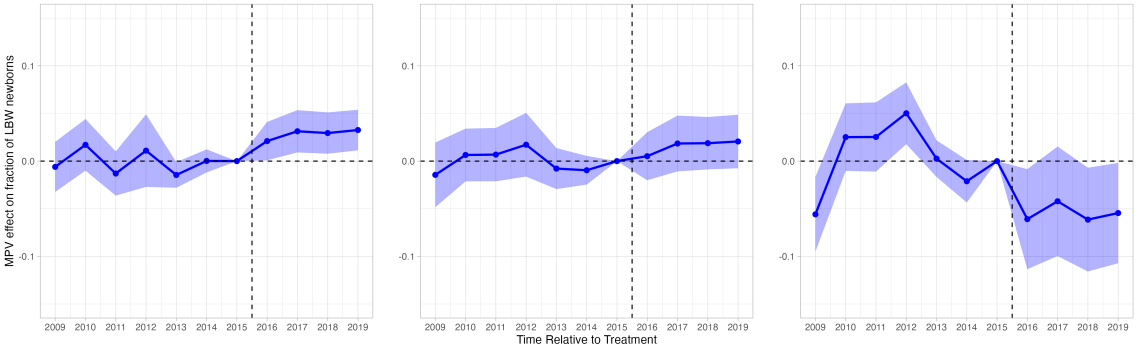
ATT ESTIMATES BY INITIAL ECONOMIC ACTIVITY

Dependent Variable	Deliveries	LBW	Weighed	Fraction LBW	Institutional Deliveries
	(1)	(2)	(3)	(4)	(5)
High	-0.0043	-0.0009	0.0082***	-0.0547**	0.0088***
	(0.0036)	(0.0020)	(0.0024)	(0.0274)	(0.0025)
Medium	0.0042**	0.0009	0.0036*	0.0158	0.0052**
	(0.0021)	(0.0009)	(0.0020)	(0.0136)	(0.0024)
Low	0.0006	0.0020***	-0.0030	0.0286***	-0.0023
	(0.0035)	(0.0006)	(0.0037)	(0.0105)	(0.0031)

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01.

By Initial Health

Heterogeneous Treatment Effects - Fraction LBW



By Initial Health

Heterogeneous Treatment Effects - Maternal Outcomes

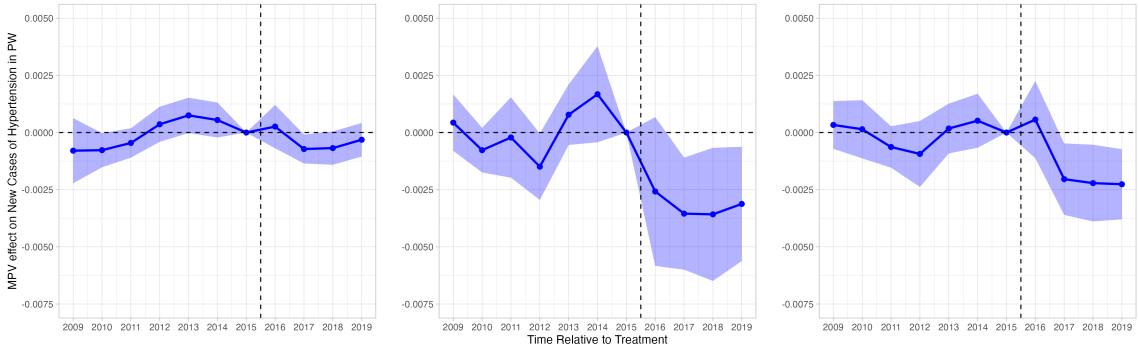
ATT ESTIMATES OF MATERNAL OUTCOMES BY ECONOMIC ACTIVITY

Dependent Variable	New Cases of Hypertension	Antenatal Care Registrations
	(1)	(2)
Full Sample	-0.0019***	0.0055
	(0.0006)	(0.0034)
High	-0.0015***	0.0212***
	(0.0004)	(0.0067)
Medium	-0.0032**	0.0020
	(0.0004)	(0.0035)
Low	0.0004	0.0129***
	(0.0004)	(0.0031)

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01.

By Initial Health

Heterogeneous Treatment Effects - New Cases of Hypertension



Antenatal Care

Summary - I

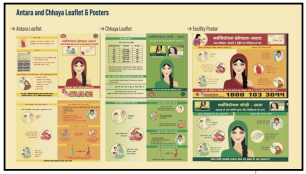
- Does the expansionary family planning policy *(MPV)* have an impact on the health outcomes of infants and mothers?
  - The estimations by the GSCM is suggestive of heterogeneous impact on outcomes of infants and mothers
  - Districts with high economic activity show a $\searrow$ in fraction of LBW infants
  - Districts with low economic activity show an $\nearrow$ in fraction LBW infants
  - New cases of hypertension among pregnant women $\searrow$ in medium- and high-economic-activity districts.

Summary - II

- Does the expansionary family planning policy *(MPV)* have an impact on the health outcomes of infants and mothers?
  - Heterogeneous effects implies that the program's effectiveness was not uniform but contingent on local baseline characteristics.
  - Aligns with diminishing marginal returns to public health interventions
    - Sufficient ⏩ health infrastructure to absorb the program and sufficient unmet fertility demand to respond
    - Effects sizes are observed to be largest where the constraints are partially but not fully binding (Dreze and Murthi (2001)).

    ,
  - The absence of a measurable impact on low birth weight despite increases in institutional deliveries is consistent with prior evidence that expanding access to maternal services improves care utilization without immediately translating into improved fetal health, which depends more on maternal nutrition, hypertension control, and long-term investments (Powell-Jackson, 2015)

What does MPV do?











Return

What does MPV do?

Date of Notification: **10th Nov 2016**

Coverage: The scheme will be applicable for all 145 High fertility districts (both rural and urban areas) as mentioned in Annexure 1.1.

Financial Package:

a) Package for service provider per insertion and ASHA per client- Rs. 150 (DO. Letter No. Y.11012/01/2013-FP, dated 27th Dec 2013)

b) Rs. 300 may be paid to the acceptors of PPIUCD to cover their incidental cost and the travel cost for two follow up visits

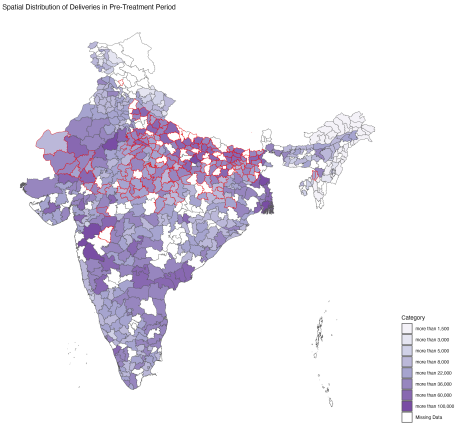
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What does MPV do?



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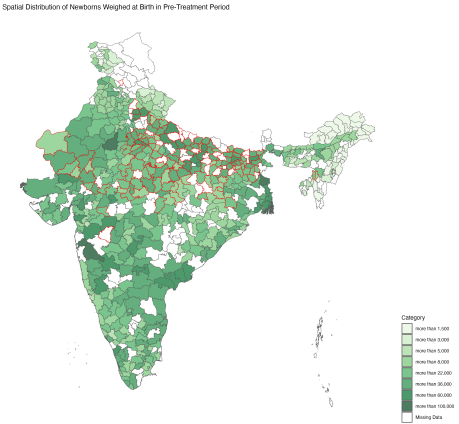
Heat Maps: Total Deliveries



This map highlights the district level variation in the average number of Total Deliveries in the pre-treatment period (2009-2010). Treated districts are outlined in red

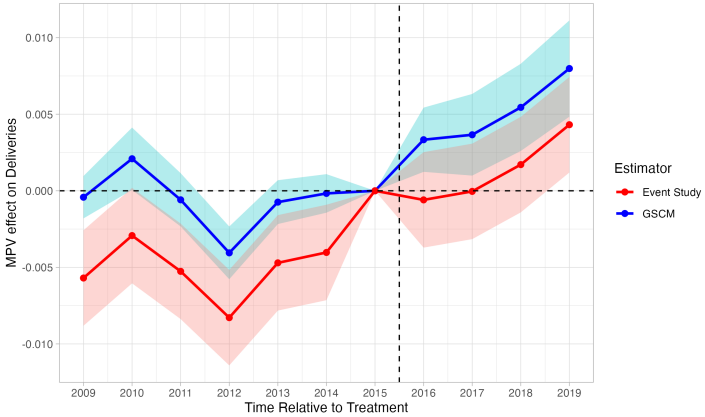
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Heat Maps: infants Weighed



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Event Study



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Summary Statistics

Predictor	Control	Treated
infant Outcomes		
Deliveries	0.0627	0.0836
Live Births	0.0629	0.0834
Weighed	0.0572	0.0726
LBW	0.0079	0.0133
Fraction LBW	0.7550	0.8120
Institutional Deliveries	0.0527	0.0629
Maternal Outcomes		
Antenatal Care Registrations	0.0856	0.1121
Hypertension in PW	0.0026	0.0040
Time Variant Covariates		
Night Light (*in ln*)	9.2579	8.9979

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Robustness Check

State	AS	CH	JR	UP	RJ	MP
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Deliveries	0.011	-0.084	0.015	0.037	-0.039	-0.007
	(0.062)	(0.275)	(0.052)	(0.080)	(0.050)	(0.023)
LBW	0.236	-0.161	0.088	0.167**	-0.082	0.075
	(0.266)	(0.324)	(0.124)	(0.082)	(0.085)	(0.104)
Live Births	0.015	0.109	0.059	0.087	-0.016	-0.012
	(0.062)	(0.092)	(0.053)	(0.053)	(0.042)	(0.023)
Infants Weighted	-0.011	0.110	0.011	-0.163	-0.021	-0.012
	(0.099)	(0.114)	(0.038)	(0.156)	(0.038)	(0.022)
Fraction LBW	0.000	0.012	0.007	0.016	-0.029	0.015
	(0.018)	(0.025)	(0.012)	(0.043)	(0.020)	(0.018)
Fraction Infants Weighed	-0.031	-0.017	-0.005	0.024	-0.023	0.002
	(0.064)	(0.029)	(0.015)	(0.044)	(0.020)	(0.011)
Antenatal Care Reg.	0.020	0.056	0.015	-0.034	0.043	-0.072
	(0.073)	(0.115)	(0.036)	(0.057)	(0.029)	(0.046)
Institutional Births	0.032	-0.119	0.069	-0.091	-0.020	0.001
	(0.092)	(0.284)	(0.064)	(0.089)	(0.044)	(0.032)
Still Births	-0.057	-0.622	0.210	0.206	0.009	0.050
	(0.258)	(0.488)	(0.262)	(0.159)	(0.170)	(0.070)
Treated Districts	2	2	9	49	13	25
Control Districts	25	12	15	13	19	24

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01.
LBW = Low-Birth Weight; **AS** = Assam; **CH** = Chhattisgarh; **JR** = Jharkhand; **UP** = Uttar Pradesh; **RJ** = Rajasthan; **MP** = Madhya Pradesh.

Heterogeneous Treatment Effects - Infant Outcomes

ATT ESTIMATES OF HEALTH OUTCOMES OF INFANTS BY INITIAL HEALTH

Dependent Variable	Deliveries	LBW	Weighed	Fraction LBW	Institutional Deliveries
Initial Health	(1)	(2)	(3)	(4)	(5)
High	-0.0026	-0.0054***	0.0060**	-0.0511	0.0055**
	(0.0025)	(0.0024)	(0.0025)	(0.0342)	(0.0025)
Medium	0.0053**	0.0005	0.0038*	0.0108	0.0072***
	(0.0023)	(0.0007)	(0.0021)	(0.0107)	(0.0025)
Low	0.0016	0.0030***	-0.0005	0.0356***	-0.0010
	(0.0026)	(0.0005)	(0.0031)	(0.0063)	(0.0030)

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01.

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Heterogeneous Treatment Effects - Maternal Outcomes

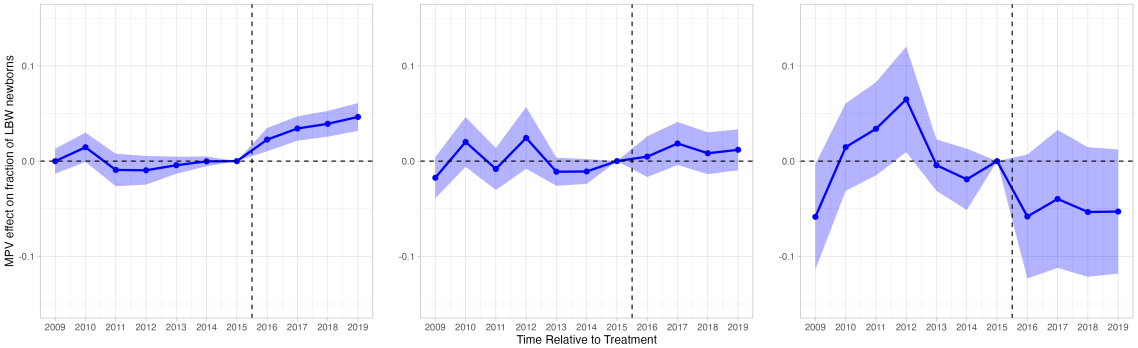
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	(1)	(2)
Full Sample	-0.0019***	0.0055
	(0.0006)	(0.0034)
High	-0.0015**	0.0212***
	(0.0007)	(0.0077)
Medium	-0.0032***	0.0020
	(0.0012)	(0.0050)
Low	-0.0004	0.0129***
	(0.0003)	(0.0032)

Notes: Standard errors are in parentheses. *p<0.1, **p<0.05, ***p<0.01.

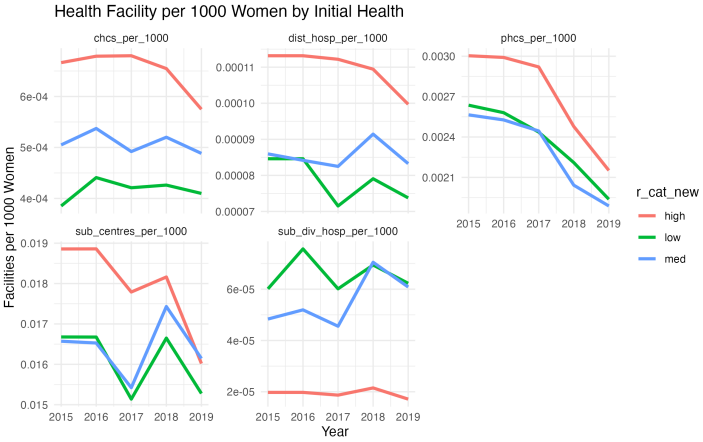
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Heterogeneous Treatment Effects - Fraction LBW



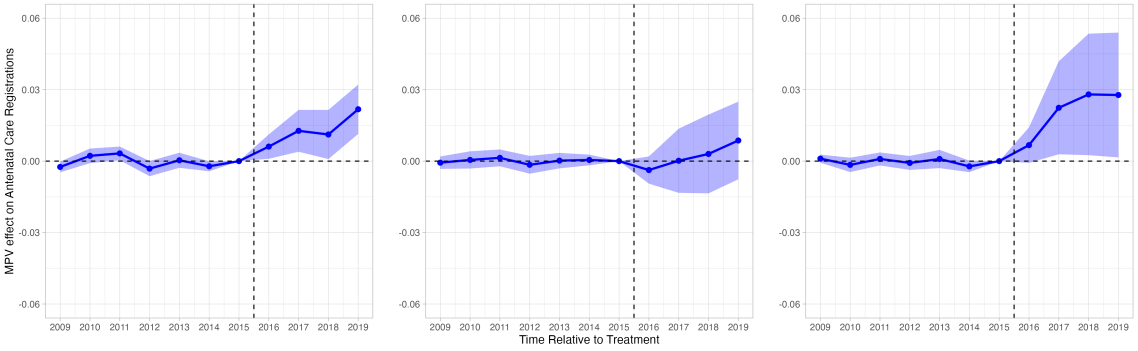
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Health Infrastructure



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Heterogeneous Treatment Effects - Antenatal Care Registrations



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